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Document Intelligence

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Validation

18 automated tests

Abstract

This project documents a layout-aware document AI stack that treats geometry as a primary signal, enabling structured field extraction, table recovery, routing, and search over heterogeneous business documents.

Project Background

Document understanding fails when text is stripped from layout. Invoices, receipts, and forms depend on where a token sits on the page, not only on the token itself, so the pipeline was designed around spatial context from the start.

Headline Result

field macro-F1 0.976; 100k docs, doc-type 100%

Scope Overview

Rendered-layout document generation with gold token boxes. Layout-aware feature pipeline and token classification. Key-value pairing, table extraction, routing, and search hooks. Evaluation across held-out documents and visual output review.

Project Context

Layout-aware document understanding for invoices, receipts, purchase orders and forms - at scale. Turn a page of tokens-with-boxes into structured, typed fields: identifiers, dates, parties, totals, and the full line-item table - then route the document and make its content searchable. Fully offline, deterministic, no GPU, no paid APIs.

Executive Summary

Background, objectives, scope, and project impact.

Objectives

- Preserve token geometry all the way through field prediction.
- Recover key-value structure and line-item tables accurately.
- Support downstream routing and search over extracted content.

Scope

- Rendered-layout document generation with gold token boxes.
- Layout-aware feature pipeline and token classification.
- Key-value pairing, table extraction, routing, and search hooks.
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This project documents a layout-aware document AI stack that treats geometry as a primary signal, enabling structured field extraction, table recovery, routing, and search over heterogeneous business documents.

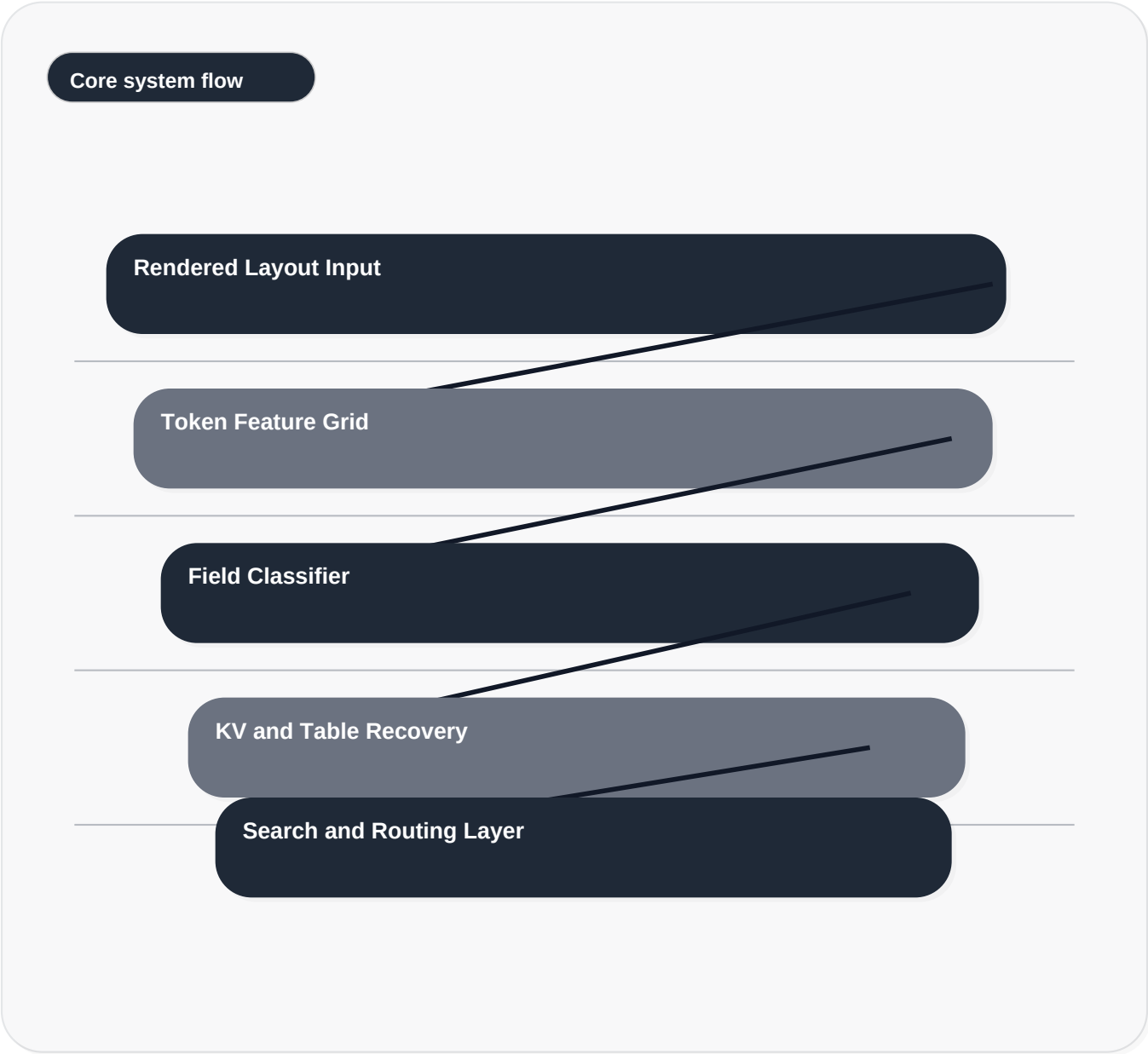
Impact

The system shows that strong field extraction and routing can be demonstrated credibly without a full OCR or vision stack, as long as the token-plus-box representation and reconstruction logic are engineered carefully.

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Solution Architecture

The pipeline stays spatial from the first token box to the recovered fields and tables, which is why the outputs stay structured.



Design Note 1

Spatial context is treated as load-bearing because field meaning depends heavily on page position.

Design Note 2

Synthetic rendered layouts keep labels exact while still reflecting realistic document structure.

Design Note 3

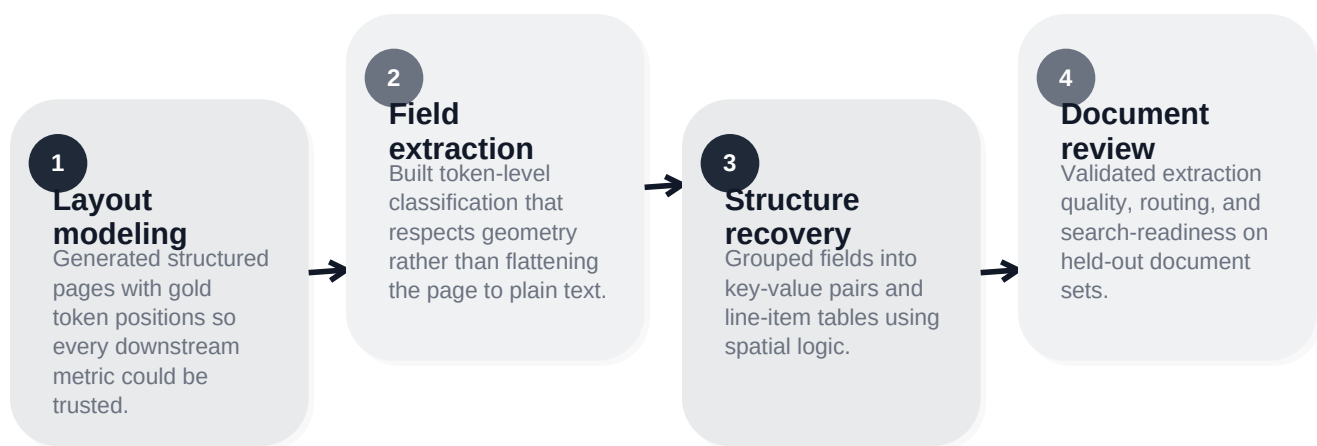
The pipeline is built for structured outputs, not just page-level classification scores.

Implementation Process

Delivery phases, engineering approach, and quality checkpoints.

Delivery Narrative

Layout-aware document understanding for invoices, receipts, purchase orders and forms - at scale. Turn a page of tokens-with-boxes into structured, typed fields: identifiers, dates, parties, totals, and the full line-item table - then route the document and make its content searchable. Fully offline, deterministic, no GPU, no paid APIs.



Quality Checkpoints

- 18 automated tests validate core behavior and edge cases.
- Benchmark files and screenshots are generated from committed project artifacts.
- The run path stays deterministic so numbers can be reproduced and reviewed directly.

Engineering Principles

- Spatial context is treated as load-bearing because field meaning depends heavily on page position.
- Synthetic rendered layouts keep labels exact while still reflecting realistic document structure.
- The pipeline is built for structured outputs, not just page-level classification scores.

Test evidence, benchmark interpretation, and screenshot review.

18 passing tests

January 27, 2025

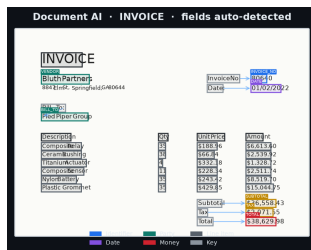
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Benchmark Evidence

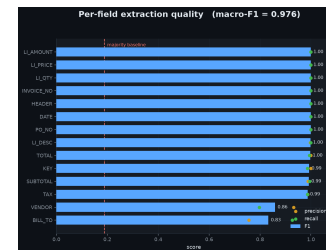
Docs	Tokens	Gen docs/s	Gen tokens/s	Infer docs/s
2,000	99,071	728	36,065	1,838
10,000	497,136	1,027	51,037	2,262
50,000	2,482,852	774	38,452	2,108
100,000	4,964,026	1,009	50,080	2,405

Result Interpretation

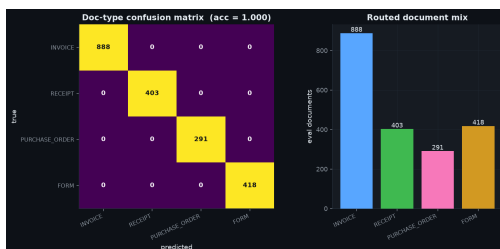
- Field-level macro-F1 is strong because the model has access to both text and layout information.
- The screenshots are useful because they show recovery of tables and key-value relationships, not only summary metrics.
- Routing quality matters operationally because extraction and downstream



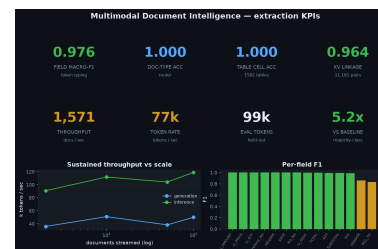
Detected fields on a rendered document (the product view)



Per-field precision / recall / F1



Document-type router: confusion matrix + routed mix



Extraction KPIs + throughput scaling

Risks And Next Steps

Operational considerations, mitigation choices, and forward path.

Risk Register

Layout edge cases Reduced by keeping geometry explicit and by separating field recovery from table reconstruction.

OCR mismatch in production Left visible as a future integration step instead of being hidden inside the prototype.

Document-type drift Addressed by routing first and preserving a modular extraction pipeline.

Next Steps

- Add scanned-document or OCR-noise simulation to stress the layout pipeline further.
- Support more document families and more complex nested table structures.
- Connect extracted entities to downstream review and retrieval workflows.

Closing Note

The project brief is intentionally written as delivered work rather than a transfer note. It documents the business problem, the engineering choices, the measured evidence, and the remaining operational path in a format suitable for portfolio review.

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Project brief date: January 27, 2025

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